

Tendon Elongation and Function After Delayed or Standard Loading of Surgically Repaired Achilles Tendon Ruptures

A Randomized Controlled Trial

Rikke Hoeffner,^{*†‡§} PT, PhD, Anne-Sofie Agergaard,^{†‡§} PT, PhD, Rene B. Svensson,^{†‡} PhD, Camilla Cullum,[§] PT, MSc, Rasmus Kramer Mikkelsen,^{||} MD, Lars Konradsen,^{||} MD, Michael Krogsgaard,^{||} MD, PhD, Mikael Boesen,[¶] MD, PhD, Michael Kjaer,^{†‡} MD, DMSc, and S. Peter Magnusson,^{†‡§} PT, DMSc
Investigation performed at Bispebjerg Hospital, Copenhagen, Denmark

Background: Achilles tendon ruptures often result in long-term functional deficits despite accelerated (standard) rehabilitation.

Purpose/Hypothesis: The purpose of this study was to investigate if delayed loading would influence functional, clinical, and structural outcomes of the muscles and tendon 1 year after a surgical repair. It was hypothesized that delaying the loading would reduce the heel-rise height deficit 1 year after Achilles tendon rupture.

Study Design: Randomized controlled trial; Level of evidence, 1.

Methods: In total, 48 patients with a surgically repaired Achilles tendon rupture were randomized to 2 groups: the standard group received the currently accepted rehabilitation, and the delayed group received the same rehabilitation except that initial loading was delayed by 6 weeks. The primary outcome was the heel-rise height difference between the injured and uninjured sides at 1 year. The secondary outcomes were (1) tendon length measured with magnetic resonance imaging, (2) muscle fascicle length and pennation angle of the gastrocnemius medialis muscle, (3) Doppler activity measured with ultrasonography, (4) Achilles tendon Total Rupture Score (ATRS), and (5) isometric muscle strength.

Results: The mean heel-rise height deficits for the standard and delayed groups were -2.2 cm and -2.1 cm, respectively ($P = .719$). The soleus part of the tendon was already elongated 1 week after surgery in both groups without a between-group difference (side-to-side difference: standard, 16.3 mm; delayed, 17.5 mm; $P = .997$) and did not change over 52 weeks. The gastrocnemius tendon length was unchanged at 1 week but elongated over time without a between-group difference (side-to-side difference at 52 weeks: standard, 10.5 mm; delayed, 13.0 mm; $P = .899$). The delayed group had less Doppler activity at 12 weeks ($P = .006$) and a better ATRS (standard, 60 points; delayed, 72 points; $P = .032$) at 52 weeks.

Conclusion: Delayed loading was not superior to standard loading in reducing the heel-rise height difference at 1 year. The data indirectly suggested reduced inflammation in the initial months and a better patient-reported outcome at 1 year in the delayed group. The soleus part of the tendon was already markedly elongated (35%) 1 week after surgery, while the length of the gastrocnemius tendon was unchanged at 1 week but was 6% elongated at 1 year. Together, these data indirectly suggest that the delayed group fared better, although this finding needs to be confirmed in future investigations.

Registration: NCT04263493 (ClinicalTrials.gov identifier).

Keywords: Achilles tendon rupture; tendon length; heel-rise height; tendon healing; surgery; tendon inflammation

The Achilles tendon is the strongest tendon in the human body.^{9,31} Yet, acute Achilles tendon rupture is a relatively common injury with an increasing incidence worldwide,^{15,22,27,39} which has reached 31 to 35/100,000 persons/

year.^{15,22,27} The injury has a male-to-female ratio of 3:1 and is frequently seen in persons 30 to 50 years of age who occasionally participate in sports.¹⁵

The optimal rehabilitation protocol after an acute Achilles tendon rupture remains undetermined. Both surgical and nonsurgical treatment approaches are commonly used, and both include a period of immobilization followed by months of rehabilitation with varying degrees of loading. Regardless of the initial surgical or nonsurgical

approach, patients often do not recover fully from an Achilles tendon rupture. For professional athletes, a rupture of the Achilles tendon can be career ending,^{23,26} and for recreational athletes, the recovery of patient-reported outcome scores is incomplete²⁰ and substantial functional deficits,²⁸ including reduced plantarflexion strength³³ and impaired heel-rise height, are reported.^{44,46} The deficit in heel-rise height appears to be related to tendon length,^{40,44,46} and it has been suggested that muscle fascicle length may also be a determinant of heel-rise height.^{4,40,46} While the maximal force-generating capacity of a muscle is determined by its physiological cross-sectional area (CSA), the excursion of the muscle is proportional to its fiber length, which would affect joint range of motion.^{25,29}

Rehabilitation has primarily focused on early mobilization after Achilles tendon rupture to minimize the loss of muscle mass and function, but these efforts have largely been unsuccessful.^{16,30} Recent endeavors have tried to avoid tendon elongation, but various rehabilitation regimens during the initial 2 months after surgery do not appear to prevent the elongation.¹² Studies indicate that tendon elongation occurs during the first 3 to 6 months.^{2,12,24,37,44} Tendon length can be assessed using methods such as ultrasonography, conventional magnetic resonance imaging (MRI), and radiography with implanted metal markers. However, the Achilles tendon is composed of 3 distinct tendons originating from the 3 separate muscles of the triceps surae,^{6,8,10} and simultaneous and precise assessment of these separate tendon components is difficult.

Standard accelerated rehabilitation does not appear to prevent the loss of muscle mass and function, and most of the elongation occurs during the first 3 to 6 months after surgery, which prompts the question if the initiation of rehabilitation should be delayed to minimize this elongation. Therefore, the purpose of the present study was to investigate if delayed loading after surgically treated Achilles tendon rupture influences the functional, clinical, and structural outcomes for muscle and tendon during the first year after surgery. We hypothesized that delaying the gradual introduction of full loading by 6 weeks in the initial 26 weeks would reduce the heel-rise height difference (primary outcome) between the injured and uninjured leg 1 year after surgery (primary endpoint).

METHODS

This trial was a single-blinded, prospective, randomized controlled superiority trial with 2 parallel arms. Ethics

approval was obtained from the regional ethics committee (No. H-19034916) and permission from the Danish Data Protection Agency was received. All participants were given oral and written information about the study and gave written informed consent. The study was preregistered at ClinicalTrials.gov (NCT04263493), and the CONSORT (Consolidated Standards of Reporting Trials) and Template for Intervention Description and Replication checklists were used as reporting guidance.

Participants

Patient recruitment was achieved in-house from the emergency clinic and the orthopaedic surgery department at University Hospital Bispebjerg-Frederiksberg Hospital, Copenhagen, Denmark. Patients 18 to 60 years of age with an acute Achilles tendon rupture were assessed for eligibility (see CONSORT flowchart in Figure 2).

Between May 2020 and September 2021, 48 patients diagnosed with a traumatic, complete Achilles tendon mid-substance rupture were included. Inclusion criteria were no contraindications for MRI, diagnosed within 14 days from injury, and able to manage transport to and from the hospital on their own. Exclusion criteria were previous Achilles tendon rupture in either limb, rerupture, other injuries affecting their lower limb function (such as late sequelae from a complex tibial or fibular fracture, stroke, etc), smoking, systemic diseases influencing healing, anticoagulation treatment, inability to follow rehabilitation or complete follow-ups, and immunosuppressive treatment including systemic corticosteroid treatment.

Randomization

Patients were randomized to the standard or delayed group. The standard group constituted the currently accepted standard regimen at the department¹² and was therefore considered the control group. A randomization procedure was performed using a computer-generated block randomization (blocks of 8) procedure.

Blinding

Neither the participants nor the health professionals (C.C.) delivering the interventions could be blinded to treatment allocation. Outcome assessors (R.H., R.B.S.) were blinded to treatment allocation, and participants were requested

*Address correspondence to Rikke Hoeffner, PT, PhD, Department of Physical and Occupational Therapy and Institute of Sports Medicine Copenhagen, Bispebjerg-Frederiksberg Hospital, Nielsine Nielsens Vej 10, Copenhagen, NV 2400, Denmark (email: rikke.hoeffner@regionh.dk) (X:@RikkeHoeffner).

[†]Institute of Sports Medicine Copenhagen, Department of Orthopaedic Surgery, University Hospital Bispebjerg-Frederiksberg, Copenhagen, Denmark.

[‡]Center for Healthy Aging, Department of Clinical Medicine, University of Copenhagen, Copenhagen, Denmark.

[§]Department of Physical and Occupational Therapy, University Hospital Bispebjerg-Frederiksberg Hospital, Copenhagen, Denmark.

^{||}Section for Sports Traumatology, Department of Orthopaedic Surgery, University Hospital Bispebjerg-Frederiksberg Hospital, Copenhagen, Denmark.

[¶]Department of Radiology, University Hospital Bispebjerg-Frederiksberg Hospital, Copenhagen, Denmark.

Submitted May 17, 2023; accepted December 6, 2023.

One or more of the authors has declared the following potential conflict of interest or source of funding: This work was supported by the Novo Nordisk Foundation (NNF180C0052371), Team Denmark, and Bispebjerg-Frederiksberg Hospital Research Foundation. AOSM checks author disclosures against the Open Payments Database (OPD). AOSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

TABLE 1
Description of the 2 Study Interventions^a

	Standard	Delayed
Plaster cast (30° of plantarflexion)	Wk 0–2	Wk 0–2
Walker orthosis	Wk 3–6	Wk 3–12
3 heel wedges; removal from	Wk 4; 1 per wk	Wk 6; 1 every second wk
No weightbearing	Wk 0–2	Wk 0–6
Partial weightbearing	From wk 3	From wk 7
Full weightbearing	From wk 7	From wk 13
ROM exercises (5 times/d)	From wk 3	From wk 3
Ergometer bike/swimming	From wk 9/9	From wk 13/15
Standing heel-rise	From wk 15	From wk 21
Jogging	6 mo after surgery	8 mo after surgery
Return to contact sport	8–9 mo after surgery	12 mo after surgery

^aROM, range of motion.

not to disclose their allocation when outcomes were assessed. Furthermore, all data were analyzed in a blinded fashion.

Sample Size Calculation

Based on previous data,¹² the heel-rise height (primary outcome) on the injured side was 76% of the uninjured side at 52 weeks, with a standard deviation of 16%. We considered a 10% deficit to be within the normal range and therefore aimed to achieve a side-to-side ratio of 90% and consequently calculated the sample size to be 21 with a power of 80% ($u_A = 76\%$; $u_B = 90\%$; $\sigma = 16\%$). To account for possible dropouts, we estimated a dropout rate of 15%; thus, a sample of 24 patients in each arm were recruited.

Surgery

Three experienced surgeons (R.K.M.) performed open repair while the patients were lying prone with their feet hanging freely at the end of the operating table. A longitudinal incision was made slightly medial to the midline at the level of the rupture. The peritenon was incised transversely and the rupture identified. The surgeon sutured the tendon using the Kessler technique with a FiberWire 2-0 and the peritenon with Vicryl No. 1. Sutures were tightened until the ankle resting angle with fully extended knee was identical to the one on the uninjured leg. A below-knee plaster cast with the ankle in 30° of plantarflexion was applied to the lower limb. The skin sutures were removed after 2 weeks without any weightbearing allowed, and the cast was replaced by an orthosis (AirSelect Standard Aircast Walker) that restricted movements of the ankle joint. The patients were assigned to 1 of 2 rehabilitation regimens (Table 1).

Outcome Measures

The primary outcome was heel-rise height difference on the injured side relative to the uninjured side at 1 year of follow-up. Secondary outcomes were heel-rise repetitions

and total work, tendon length, muscle CSA, and fat fraction for the soleus and gastrocnemii muscles; Achilles tendon Total Rupture Score (ATRS); physical activity scale; a 5-point Likert scale for satisfaction; medial gastrocnemius muscle fascicle length, thickness, and pennation angle; power Doppler flow in the Achilles tendon; and plantarflexion isometric muscle strength.

Heel-Rise Test

Heel-rise muscle function (height) and endurance were evaluated using a standardized heel-rise test at 26 and 52 weeks.^{43,44} The patient completed a 5-minute warm-up on an exercise bike followed by 10 bilateral heel rises. The patient was standing on a box with the ankle in neutral position and his or her fingertips placed at shoulder height against the wall for balance. The heel-rise test was performed barefooted on 1 leg at a time and the uninjured leg was tested first. The test was performed with a cadence at 30 rises per minute, ensured by a metronome. The MuscleLab (Ergotest Innovation AS) computer software and a linear encoder sensor were used for evaluation. The linear encoder was attached to the heel and the number of heel rises and the height of each heel rise were measured. The patients were instructed to go as high as possible and perform as many heel rises as possible. No verbal encouragement was provided. The test was terminated when the participant stopped, was unable to maintain the frequency, or could not perform the heel rise without compensatory strategies (eg, by bending the knees or leaning too much against the wall). Maximum height, repetitions, and total work (body weight multiplied by total distance) were used for the data analysis.

Magnetic Resonance Imaging

An MRI examination (Version 5.6.1.2; Ingenia Ambition 1.5T scanner software; Philips) was performed on all patients at weeks 1, 12, 26, and 52. The patients were scanned in a supine position with their legs close together

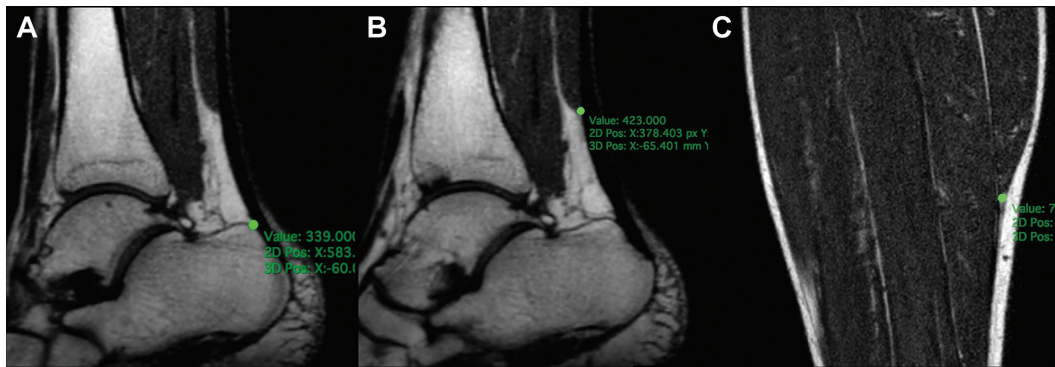


Figure 1. Landmarks used to measure the Achilles tendon length. (A) The most proximal insertion of the “free” Achilles tendon on the calcaneus. (B) The most distal part of the myotendinous junction (MTJ) between the soleus muscle and Achilles tendon. (C) The most distal part of the MTJ between the medial gastrocnemius muscle and Achilles tendon.

and their feet placed against a plastic foot plate. The injured leg was placed in a standard position using the same plaster cast that was applied just after their surgery, and a rubber band around both forefeet was used to ensure a consistent imaging position. A coronal isotropic 3-dimensional (3D) T1-weighted sequence (voxel size, $0.7 \times 0.7 \times 0.7 \text{ mm}^3$) of both ankle and calf was used to measure the tendon length, including the soleus and gastrocnemius tendon parts, and the CSA. An axial 6-point Dixon sequence was applied to assess the free fat fraction of the muscles. The Horos open-source medical image viewer (Version 3.3.6) was used for tendon length and CSA, and Fiji/ImageJ2 (Version 2.3.0) was used for Dixon free fat fraction analysis. For the tendon length measurements, 3 landmark points in 3D were defined (Figure 1) in each lower leg as previously outlined.⁴⁶ Through the use of this approach it has been shown that there is no systematic bias on repeated measures and that the correlation coefficient and typical error percentage range from 0.94 to 0.98 and 3.15% to 5.1%, respectively.⁴⁶ The CSAs of the soleus and the medial and lateral gastrocnemius muscles were measured manually at one-third of the length of the triceps surae muscle. Both the injured and the uninjured legs were analyzed.

Ultrasonography

The muscle fascicle length, pennation angle, and thickness of the medial gastrocnemius muscle were measured at weeks 2, 12, 26, and 52 with B-mode ultrasonography (HI VISION Ascendus; Hitachi Medical Systems). The ultrasonography examinations were performed with the patient lying prone with the knees extended in 2 positions: position 1 (free, at week 2), with the feet and ankles lying free of the bed; and position 2 (supported, at weeks 12, 26, and 52), with the feet and ankles supported in plantarflexion by a foam block to reproduce the same degree of plantarflexion that was measured at week 2 on the injured side in the free position. Note that the uninjured side was also placed at the angle corresponding to the injured side at 2 weeks to enable comparison at equal plantarflexion angles

between legs. A protocol based on previously described techniques was used.^{7,21} Images were obtained at 3 locations: medial, central, and lateral at the midmuscle belly.

Power Doppler flow in the Achilles tendon was recorded at weeks 12, 26, and 52. In the area with the highest visible power Doppler activity in the sagittal plane, two 5-second sine loops (20 frames) were recorded. Fiji/ImageJ2 was used for quantitative analysis. To analyze fascicle length, pennation angle, and muscle thickness, a previously published plug-in (Simple Muscle Architecture Analysis) was used to minimize assessor influence.⁴¹ Another in-house plug-in was used to quantify power Doppler flow by automatically detecting the frame with the most Doppler signal within the tendon after manual removal of noise artifacts. Analyses were performed by persons (R.H., R.B.S.) blinded to intervention allocation and side.

Plantarflexion Torque

Patients were seated (Biodex Multi-joint System 4 Pro; Biodex Medical Systems) with the hip flexed to 70° and the knee in full extension (0°). Warm-up consisted of 3 isometric plantar contractions (approximately 50%, 80%, and 100%). The patient performed a series of maximal isometric plantar contraction (3-5 repetitions) at -10° , 0° , 10° , and 20° as described in a previous study.³³ There was a rest period of 30 seconds between each contraction and a 60-second rest period between the different angles. Both the injured and the uninjured legs were tested, starting with the uninjured one. The maximal peak torque at each angle was used for data analysis. This test was only conducted at the 12-month follow-up.

Patient-Reported Outcome Measures

Patient symptoms, function, and treatment satisfaction were evaluated using the ATRS, the physical activity scale, and a 5-point Likert scale for satisfaction. The ATRS is a patient-reported instrument for measuring the outcome related to symptoms and physical activity after treatment in patients with a total Achilles tendon rupture.³⁵ A score

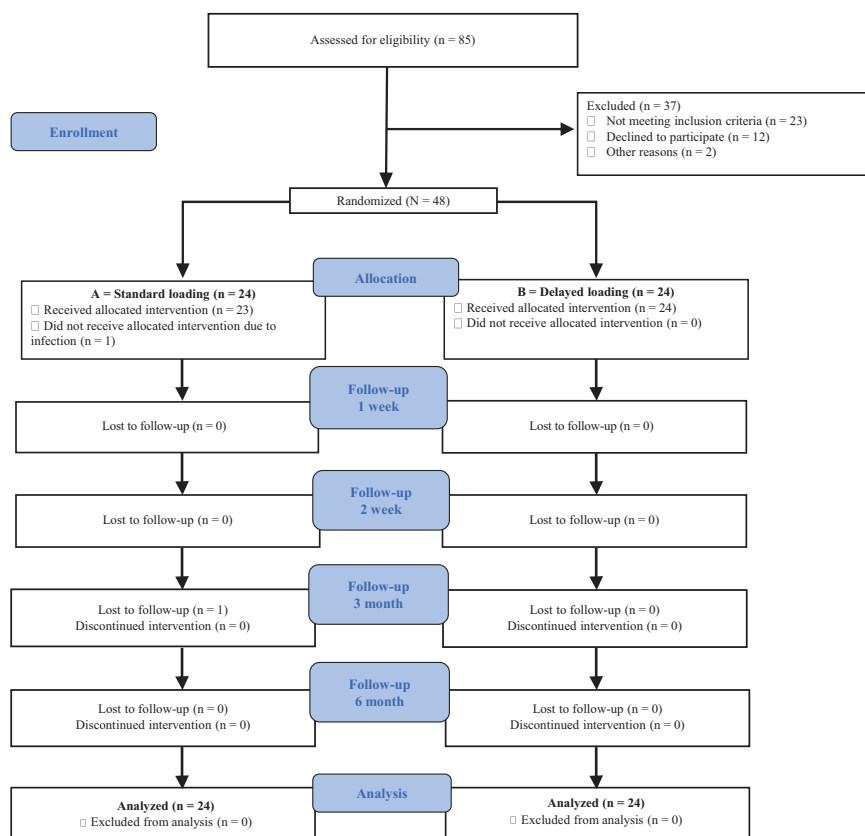


Figure 2. CONSORT (Consolidated Standards of Reporting Trials) flowchart.

of 100 points indicates full function and no pain, with lower scores indicating reduced function. The Danish version of the ATRS questionnaire has been validated by classic test theory methods.¹⁴ The physical activity scale questionnaire is a self-report instrument for measuring physical activity based on the intensity of specific activities expressed in metabolic equivalent.¹ Both the ATRS and the physical activity scale questionnaires were completed at baseline (as recall 1 week before injury) and at 52 weeks of follow-up. At 52 weeks of follow-up, a 5-point Likert scale for satisfaction was completed with the following 2 questions: (1) How satisfied are you with the function of your operated leg in everyday life? (2) How satisfied are you with the function of your operated leg in relation to sports/activities that include jumping and running?

Statistical Analysis

All data were analyzed using GraphPad Prism Version 9.4.1 following a predefined statistical analysis plan. Results are reported as mean $\pm$ SEM unless otherwise noted. Difference values refer to the injured minus the uninjured side in the different outcome measures unless otherwise noted. Patient baseline characteristics were analyzed by use of the unpaired Student *t* test. A 2-way mixed model (group $\times$ time) that included all time points was used to analyze the difference values in the outcome

parameters. For isometric strength measurements, only a single time point (52 weeks) was measured, and therefore a 2-way mixed model (group $\times$ ankle angle) was used to analyze the percentage difference values. All analyses were performed as intention to treat. The significance level for all tests was set to $P \leq .05$.

RESULTS

Of 85 participants assessed for eligibility, 37 were excluded (Figure 2). There were no dropouts in the groups, but 1 participant in the standard group got a deep infection and was treated with silver ions and consequently excluded from the MRI and ultrasound examinations at the 3-month follow-up examination. There were no reruptures.

The baseline characteristics of the patients are presented in Table 2.

Adverse Events and Complications

In the standard group, there were 2 deep infections reported 11 weeks after surgery, 1 deep vein thrombosis that occurred 10 days after surgery, and 2 patients who complained of nerve disturbance/pain under or on the lateral aspect of the foot at 26 weeks, but this was not present at 52 weeks. There were no reported complications in the delayed group.

TABLE 2
Baseline Patient Characteristics^a

	Standard, n = 24	Delayed, n = 24
Male sex, n (%)	17 (71)	18 (75)
Age, y	36 ± 10	37 ± 9
Height, cm	179 ± 8	179 ± 11
BMI	25 ± 3	26 ± 3
Injured side, right/left, n	10/14	12/12
ATRS (recall at 1 wk), points	99 ± 3	99 ± 3
PAS (recall at 1 wk), MET-h	43.6 ± 7	47.6 ± 8

^aData are presented as mean ± SD unless otherwise indicated. ATRS, Achilles tendon Total Rupture Score; BMI, body mass index; MET-h, metabolic equivalent of task hours; PAS, physical activity scale.

Heel-Rise Test

There was an effect of time ($P < .0001$), but no group or interaction effect with respect to heel-rise height deficit at 26 ($P = .682$) or 52 (standard, -2.2 cm; delayed -2.1 cm; $P = .719$) weeks (Figure 3; Appendix Table A1, available in the online version of this article). There was an effect of time ($P < .0001$), but no group or interaction effect with respect to heel-rise repetition deficit at 26 (standard, -8; delayed, -12; $P = .099$) and 52 (standard, -2; delayed, -5; $P = .199$) weeks. There was an effect of time ($P < .0001$), but no group or interaction effect with respect to heel-rise work deficit (body weight times total distance) at 26 (standard, -957 J; delayed, -1343 J; $P = .065$) or 52 (standard, -517 J; delayed, -720 J; $P = .279$) weeks.

Tendon Length

There was no significant group or interaction effect, but a main effect of time for both the soleus part and the gastrocnemius part (Figure 4, A and B; Appendix Tables A2 and A3, available online). The side-to-side difference in soleus tendon length corresponded to a 35% (standard, 16.3 mm; delayed, 17.5 mm; $P = .0997$) elongation at week 1, 30% at week 12, 41% at week 26, and 42% at week 52. The side-to-side difference in gastrocnemius tendon length corresponded to a 1% elongation on the injured side at week 1, 4% at week 12, 5% at week 26, and 6% (standard, 10.5 mm; delayed, 13.0 mm; $P = .899$) at week 52.

Power Doppler Activity

An interaction was observed between time and group ($P < .0001$). At week 12, the delayed group had less Doppler area than the standard group ($P = .006$), while there was no difference at 26 and 52 weeks (Figure 5; Appendix Table A4, available online). In the standard group, there was a significant decrease in Doppler area over time from 12 to 26 weeks ($P < .0001$) and from 26 to 52 weeks ($P = .0006$). In the delayed group, there was no change in Doppler area from 12 to 26 weeks but a significant decrease from 26 to 52 weeks ($P = .0003$).

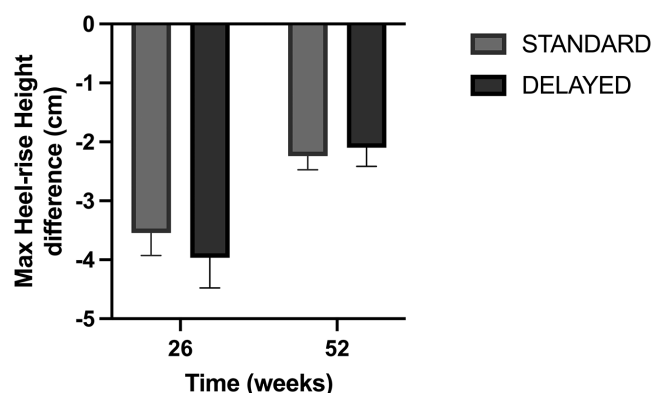


Figure 3. Heel-rise height difference between injured and uninjured sides in the standard and delayed groups at 26 and 52 weeks. Data are presented as mean ± SEM.

Muscle Dimensions

There was no group or interaction effect, but an effect of time ($P < .0001$) with reduced gastrocnemius medialis muscle thickness after 26 weeks, fascicle shortening after 26 weeks, and increased pennation after 26 weeks (Table 3; Appendix Figures A3 and A4, available online).

For CSA of the soleus, gastrocnemius medialis, and gastrocnemius lateralis muscles, there was no group or interaction effect, but an effect of time ($P < .0001$) (Table 3; Appendix Figure A1, available online). At all time points, there was a difference ($P < .0001$) between the injured and the uninjured side for the soleus, gastrocnemius medialis, and gastrocnemius lateralis muscles. The soleus muscle CSA was 6% smaller on the injured side at week 1, 21% at week 12, 21% at week 26, and 20% at week 52. The gastrocnemius medialis muscle CSA was 16% smaller on the injured side at week 1, 27% at week 12, 14% at week 26, and 13% at week 52. The gastrocnemius lateralis muscle CSA was 18% smaller on the injured side at week 1, 26% at week 12, 10% at week 26, and 4% at week 52.

For free fat fraction, there was no group or interaction effect, but an effect of time ($P < .0001$) (Table 3; Appendix Figure A2, available online). At all time points, there was a significant difference ($P < .0001$) between the injured and the uninjured side for the soleus, gastrocnemius medialis, and gastrocnemius lateralis muscles. The soleus free fat fraction was 15% greater on the injured side at week 1, 46% at week 12, 58% at week 26, and 50% at week 52. The gastrocnemius medialis free fat fraction was 21% on the injured side at week 1, 29% at week 12, 31% at week 26, and 25% at week 52. The gastrocnemius lateralis free fat fraction was 8% greater on the injured side at week 1, 20% at week 12, 16% at week 26, and 18% at week 52.

Plantarflexion Torque

No interaction was observed for the maximal peak torque between group and ankle angle. There was a difference between the injured and uninjured sides ($P < .0001$) and

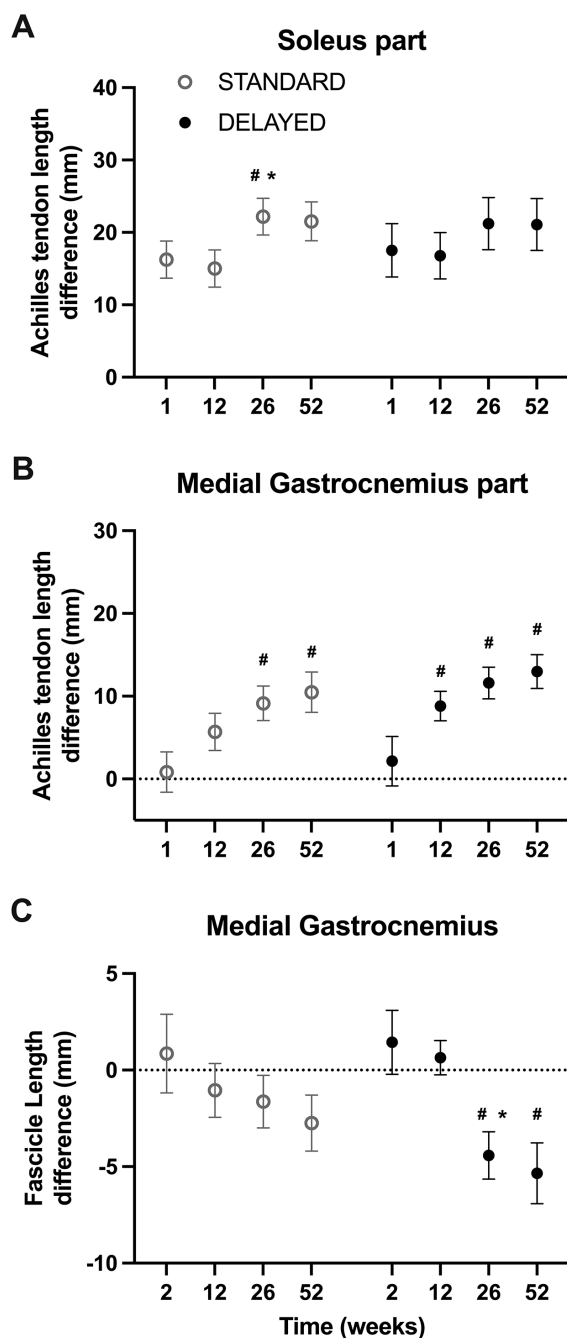


Figure 4. The Achilles tendon length difference between the injured and uninjured sides at different time points in the standard and delayed groups. (A) Soleus part. (B) Medial gastrocnemius part. (C) Fascicle length (medial gastrocnemius muscle) difference between the injured and uninjured sides at different time points. *Different from preceding time point. #Different from 1 week. Data are presented as mean $\pm$ SEM.

between the different angles ($P < .0001$): a lower torque on the injured side and decreasing torque with greater plantarflexion was seen, but no difference between the 2 groups (Appendix Table A5, available online).

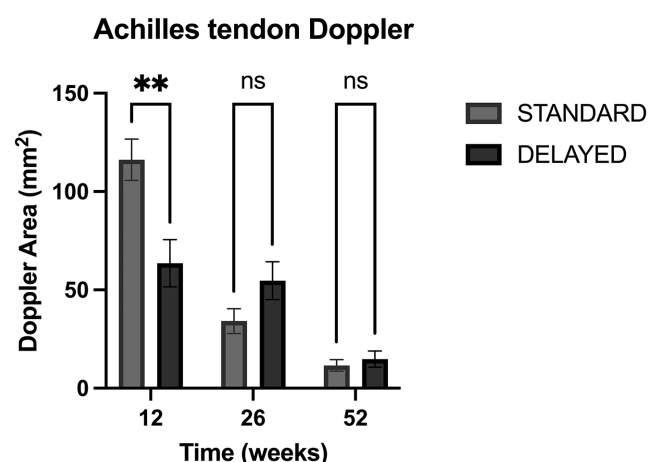


Figure 5. Area (mm²) with power Doppler in the injured Achilles tendon measured at different time points. ** $P < .01$ Data are presented as mean $\pm$ SEM. ns, not significant.

Patient-Reported Outcome Measures

Regarding ATRS, an interaction was observed between time and group ($P = .021$) and a significant effect of time ($P < .0001$). At preinjury (reported as recall 1 week before injury) there was no difference between the standard and delayed groups (Appendix Table A6, available online), but at 52 weeks there was a difference between the standard and delayed groups, with a mean difference of 12.5 ± 5.7 (95% CI, 3.3-21.8; $P = .053$). There was no significant group, time, or interaction effect with respect to the physical activity scale. The metabolic equivalent-hours for the physical activity scale preinjury was 43.64 ± 1.5 for the standard group and 47.64 ± 1.5 for the delayed group ($P = .0909$), and no changes were seen in metabolic equivalent-hours at 52 weeks ($P = .9429$) (Appendix Table A6, available online). The 5-point Likert scale (only measured a 52 weeks) for satisfaction in relation to everyday living was 4.3 points for the standard group and 4.6 points for the delayed group ($P = .076$), and correspondingly in relation to sports/activities, 2.8 points for the standard group and 3.4 points for the delayed group ($P = .081$).

DISCUSSION

The present randomized controlled trial investigated if delayed loading after surgically treated acute Achilles tendon rupture influenced the clinical outcome, function, and muscle and tendon structure after 1 year. The data suggest that a delayed loading rehabilitation protocol was not superior to standard loading with respect to the heel-rise height difference (primary outcome) at 1 year after surgery. However, the delayed group had less Doppler activity in the initial 12 weeks and a better ATRS at 52 weeks. Interestingly, the soleus tendon was already longer (35%) on the injured side 1 week after surgery, and only elongated marginally over 52 weeks, while the gastrocnemius

TABLE 3
Muscle Architecture^a

	All (uninjured) ^b	Standard (injured) ^c				Delayed (injured) ^c			
		1 wk	12 wk	26 wk	52 wk	1 wk	12 wk	26 wk	52 wk
Thickness, mm	19.3 ± 0.7	16.8 ± 0.6	16.1 ± 0.6	17.7 ± 0.6	18.1 ± 0.6	17.4 ± 0.6	16.4 ± 0.5	17.8 ± 0.5	18.1 ± 0.5
Fascicle length, mm	42.9 ± 1.6	44.3 ± 0.6	42.9 ± 1.3	40.3 ± 1.4	39.6 ± 1.7	44.1 ± 1.6	43.3 ± 1.3	38.8 ± 1.2	36.9 ± 0.9
Pennation angle, deg	29.3 ± 1.1	25.2 ± 0.9	24.9 ± 0.6	29.4 ± 0.9	30.7 ± 1.1	26.8 ± 0.8	26.1 ± 0.8	31.6 ± 1.0	32.5 ± 0.9
CSA, mm ²									
SOL	3035 ± 134	2775 ± 177	2369 ± 136	2422 ± 117	1280 ± 137	2967 ± 129	2339 ± 111	2521 ± 116	2649 ± 119
GM	1547 ± 89	1273 ± 81	1142 ± 62	1301 ± 80	1399 ± 83	1308 ± 84	1125 ± 55	1349 ± 73	1374 ± 77
GL	721 ± 61	548 ± 48	483 ± 32	618 ± 50	670 ± 53	663 ± 56	564 ± 44	658 ± 55	754 ± 66
Fat fraction, %									
SOL	7.6 ± 0.6	8.6 ± 0.6	10.1 ± 0.6	12.0 ± 1.0	11.0 ± 1.0	8.7 ± 0.6	12.1 ± 1.0	13.0 ± 1.1	12.0 ± 1.3
GM	8.4 ± 0.5	9.8 ± 0.6	10.6 ± 0.6	10.9 ± 0.6	10.4 ± 0.6	9.8 ± 0.6	11.9 ± 1.0	11.8 ± 0.8	10.6 ± 0.9
GL	8.5 ± 0.6	8.9 ± 0.4	10.1 ± 0.6	9.5 ± 0.5	9.0 ± 0.8	9.5 ± 0.6	10.7 ± 0.8	10.2 ± 0.8	10.2 ± 0.8

^aData are presented as mean ± SEM. Muscle thickness, muscle fascicle length, and pennation angle of the gastrocnemius medialis muscle were measured by ultrasound. The cross-sectional areas were measured by magnetic resonance imaging. CSA, cross-sectional area; GL, gastrocnemius lateralis muscle; GM, gastrocnemius medialis muscle; SOL, soleus muscle.

^bMean of all uninjured sides at all time points.

^cMean of injured sides in each group, standard and delayed, at 1, 12, 26, and 52 weeks (ultrasound measures at week 2).

tendon length was unchanged at 1 week but elongated (6%) over 52 weeks.

There were no reruptures in either group, and there were 4 complications in the standard group: 1 deep infection and 1 deep vein thrombosis, and 2 patients experienced nerve sensory disturbance symptoms 26 weeks after surgery, which all resolved at 52 weeks. Notably, there were no reported complications in the delayed group.

The loading initiation did not influence the heel-rise height (Figure 3), which is similar to results of previous studies.^{2,12,34,36} The deficit in heel-rise height may be influenced by tendon and muscle fascicle length more than strength,^{4,40,44,46} because the total excursion of the muscle (joint range of motion) roughly corresponds to its fiber length.^{25,29} In the current study, the soleus and gastrocnemius tendons were elongated (Figure 4, A and B) and the gastrocnemius muscle fascicles were shortened (Figure 4C) after 52 weeks. If the reduction in muscle fascicle length is accompanied by an adaptation of sarcomeres in series remains unknown.

The soleus muscle had retracted substantially, while the gastrocnemius muscles had a length comparable with the muscles of the contralateral side 1 week postoperatively. This retraction may have occurred during the first week postoperatively, but it is unlikely because there had been no weightbearing load on the tendon before the MRI examination took place. It seems more likely that the normal length of the soleus tendon part of the Achilles tendon was not achieved by surgery, and to the best of our knowledge, this is the first documentation of a differential tendon elongation after a surgical repair.

Precisely assessing tendon length can be challenging. Using tantalum beads coupled with radiographs precludes differential tendon assessment between the soleus muscle and gastrocnemius part; ultrasound also makes differential tendon assessment difficult, especially shortly after

surgery, whereas precision of repeated discrete sagittal images on conventional MRI can be difficult. We used 3D T1-weighted isotopic MRI, which permits reliable⁴⁶ reconstruction of the imaging data to measure differential tendon length, including in the early stage after surgery. Therefore, it is difficult to compare the elongation of the tendon in the present study to that of previous studies using other methods; however, studies employing ultrasound or MRI have reported tendon elongations between 11 and 22 mm,^{3,19,44,45} which is comparable to the present data (soleus muscle, 21 mm; gastrocnemius muscle, 12 mm). Moreover, the elongation pattern over time of the gastrocnemius tendon resembles that of previous data.¹²

The healing response after surgical repair of an Achilles tendon rupture includes an inflammatory phase (1–2 weeks), a proliferative or reparative phase (1–2 months), and the final remodeling phase (up to 1 year).^{13,42} Unsurprisingly, there was increased vascularization in the injured tendon compared with the uninjured side (Figure 5) that declined over the 52 weeks, which corresponds well with a previous report of vascularization and metabolic activity after surgical repair.¹³ However, the delayed group had less vascularization in the injured tendon than the standard group at 12 weeks. Metabolic activity appears inversely related to patient-reported outcome at 6 months,¹³ and therefore it may be that standard loading amplified the normal reparative response in tendon in this period. The implication of this observation is unclear and needs to be considered in future studies.

There can be a substantial side-to-side strength deficit years after an Achilles tendon rupture,²⁰ which the present data corroborate (Table A5). This strength loss likely relates to the loss of muscle mass (Table 3). Moreover, the deficit appears amplified in the more plantarflexed position (approximately 23%), which is in accordance with the previously reported end-range plantarflexion

weakness of 28% to 34%.^{32,33,38} The gastrocnemius medialis (16%) and lateralis (18%) muscles showed pronounced atrophy already 1 week after surgery, while that of the soleus muscle was somewhat less (6%). All 3 muscles continued to lose muscle mass until week 12. Interestingly, the gastrocnemii muscles only displayed 13% and 4% deficits, respectively, by week 52. In contrast, the soleus muscle showed a side-to-side difference of 20%, which is consistent with earlier studies.^{12,18} The reason for this persistent and substantial muscle mass deficit of the soleus muscle remains unknown, but it may relate to the elongation of the soleus part of the tendon, which may affect muscular activation.

We showed that there is a significant lipid deposition in the triceps surae muscles (Table 3). The increase in fat deposition in the current study is likely an effect of loss of contractile activity.^{5,17} It has previously been shown that fat infiltration is present in the triceps surae muscle group years after a tendon rupture.¹¹ Here we corroborate those findings and show that the fat infiltration is muscle specific: after 52 weeks, the free fat fractions were 50%, 25%, and 18% greater on the injured side than on the uninjured side in the soleus, medial gastrocnemius, and lateral gastrocnemius muscles, respectively.

The ATRS is the patient-reported outcome of symptoms and physical activity after a total Achilles tendon rupture,³⁵ and in the present study the delayed group scored higher (72 points) than the standard group (60 points) at week 52, which represents a clinically meaningful difference.³⁵ Although the difference was statistically significant and represented a clinically meaningful difference, it should be pointed out that the confidence interval was rather wide, which is largely a function of the sample size. At the same time, the greater ATRS in the delayed group was coupled with a tendency toward greater satisfaction in relation to everyday living (Likert scale, $P = .076$) and sports activities (Likert scale, $P = .081$) compared with the standard group. Together, these data indirectly suggest that the delayed group fared better, although this needs to be confirmed in future investigations.

Unfortunately, we were unable to obtain soleus muscle fascicle length with sufficient image quality, and lateral gastrocnemius tendon length was also not measured. The patients in both groups received instructions and treatment from the same physical therapist at weeks 2, 6, 12, and 18 and were then referred to additional rehabilitation in the municipality, but the content of and compliance with this rehabilitation were not recorded.

CONCLUSION

Delayed loading was not superior compared with standard loading in the primary endpoint of reducing the heel-rise height difference at 1 year after acute surgery for Achilles tendon rupture. Both groups were unable to completely recover with respect to ATRS, muscle strength, and heel-rise function. Furthermore, 3D MRI of the Achilles tendon length showed no group differences, but all patients, regardless of rehabilitation regimen, had a significant

35% soleus tendon elongation compared with the contralateral side at 1 week that elongated further to 42% over 52 weeks after surgery, while the gastrocnemius tendon length was intact at 1 week but elongated 6% during 52 weeks. Moreover, the delayed group had less Doppler activity in the initial 12 weeks, which could be contributed to differences in rehabilitation, and they had a significantly greater ATRS at 52 weeks, suggesting a better patient-reported outcome in this group, which must be confirmed in future trials.

REFERENCES

1. Aadahl M, Jørgensen T. Validation of a new self-report instrument for measuring physical activity. *Med Sci Sports Exerc.* 2003;35(7):1196-1202.
2. Aufwerber S, Edman G, Gravare Silbernagel K, Ackermann PW. Changes in tendon elongation and muscle atrophy over time after Achilles tendon rupture repair: a prospective cohort study on the effects of early functional mobilization. *Am J Sports Med.* 2020;48(13):3296-3305.
3. Barfod KW, Hansen MS, Hölmich P, Kristensen MT, Troelsen A. Efficacy of early controlled motion of the ankle compared with immobilisation in non-operative treatment of patients with an acute Achilles tendon rupture: an assessor-blinded, randomised controlled trial. *Br J Sports Med.* 2020;54(12):719-724.
4. Baxter JR, Hulfish TJ, Chao W. Functional deficits may be explained by plantarflexor remodeling following Achilles tendon rupture repair: preliminary findings. *J Biomech.* 2018;79:238-242.
5. Bayer ML, Hoegberg-Kalish M, Svensson RB, et al. Chronic sequelae after muscle strain injuries: influence of heavy resistance training on functional and structural characteristics in a randomized controlled trial. *Am J Sports Med.* 2021;49(10):2783-2794.
6. Bojsen-Møller J, Magnusson SP. Heterogeneous loading of the human Achilles tendon in vivo. *Exerc Sport Sci Rev.* 2015;43(4):190-197.
7. Bolsterlee B, Gandevia SC, Herbert RD. Ultrasound imaging of the human medial gastrocnemius muscle: how to orient the transducer so that muscle fascicles lie in the image plane. *J Biomech.* 2016;49(7):1002-1008.
8. Cummins EJ, Anson BJ, Carr BW, et al. The structure of the calcaneal tendon (of Achilles) in relation to orthopedic surgery, with additional observations on the plantaris muscle. *Surg Gynecol Obstet.* 1946;83:107-116.
9. Doral MN, Alam M, Bozkurt M, et al. Functional anatomy of the Achilles tendon. *Knee Surg Sports Traumatol Arthrosc.* 2010;18(5):638-643.
10. Edama M, Kubo M, Onishi H, et al. The twisted structure of the human Achilles tendon. *Scand J Med Sci Sports.* 2015;25(5):e497-e503.
11. Eken G, Misir A, Tangay C, et al. Effect of muscle atrophy and fatty infiltration on mid-term clinical, and functional outcomes after Achilles tendon repair. *Foot Ankle Surg.* 2021;27(7):730-735.
12. Eliasson P, Agergaard AS, Couppe C, et al. The ruptured Achilles tendon elongates for 6 months after surgical repair regardless of early or late weightbearing in combination with ankle mobilization: a randomized clinical trial. *Am J Sports Med.* 2018;46(10):2492-2502.
13. Eliasson P, Couppe C, Lonsdale M, et al. Ruptured human Achilles tendon has elevated metabolic activity up to 1 year after repair. *Eur J Nucl Med Mol Imaging.* 2016;43(10):1868-1877.
14. Ganestam A, Barfod K, Klit J, Troelsen A. Validity and reliability of the Achilles tendon Total Rupture Score. *J Foot Ankle Surg.* 2013;52(6):736-739.
15. Ganestam A, Kallermose T, Troelsen A, Barfod KW. Increasing incidence of acute Achilles tendon rupture and a noticeable decline in surgical treatment from 1994 to 2013. A nationwide registry study of 33,160 patients. *Knee Surg Sports Traumatol Arthrosc.* 2016;24(12):3730-3737.
16. Ghaddaf AA, Alomari MS, Alsharef JF, Alakkas E, Alshehri MS. Early versus late weightbearing in conservative management of acute

- Achilles tendon rupture: a systematic review and meta-analysis of randomized controlled trials. *Injury*. 2022;53(4):1543-1551.
17. Gorgey AS, Mather KJ, Cupp HR, Gater DR. Effects of resistance training on adiposity and metabolism after spinal cord injury. *Med Sci Sports Exerc*. 2012;44(1):165-174.
 18. Heikkinen J, Lantto I, Flinkkila T, et al. Soleus atrophy is common after the nonsurgical treatment of acute Achilles tendon ruptures: a randomized clinical trial comparing surgical and nonsurgical functional treatments. *Am J Sports Med*. 2017;45(6):1395-1404.
 19. Heikkinen J, Lantto I, Piihonen J, et al. Tendon length, calf muscle atrophy, and strength deficit after acute Achilles tendon rupture: long-term follow-up of patients in a previous study. *J Bone Joint Surg Am*. 2017;99(18):1509-1515.
 20. Hoeffner R, Svensson R, Bjerregaard N, Kjaer M, Magnusson S. Persistent deficits after an Achilles tendon rupture: a narrative review. *Transl Sports Med*. 2022;2022:7445398.
 21. Hullfish TJ, O'Connor KM, Baxter JR. Gastrocnemius fascicles are shorter and more pennate throughout the first month following acute Achilles tendon rupture. *PeerJ*. 2019;7:e6788.
 22. Huttunen TT, Kannus P, Rolf C, Felländer-Tsai L, Mattila VM. Acute Achilles tendon ruptures: incidence of injury and surgery in Sweden between 2001 and 2012. *Am J Sports Med*. 2014;42(10):2419-2423.
 23. Johns W, Walley KC, Seedat R, et al. Career outlook and performance of professional athletes after Achilles tendon rupture: a systematic review. *Foot Ankle Int*. 2021;42(4):495-509.
 24. Kangas J, Pajala A, Ohtonen P, Leppilahti J. Achilles tendon elongation after rupture repair: a randomized comparison of 2 postoperative regimens. *Am J Sports Med*. 2007;35(1):59-64.
 25. Khair RM, Stenroth L, Cronin NJ, et al. Muscle-tendon morphomechanical properties of non-surgically treated Achilles tendon 1-year post-rupture. *Clin Biomech (Bristol, Avon)*. 2022;92:105568.
 26. LaPrade CM, Chona DV, Cinque ME, et al. Return-to-play and performance after operative treatment of Achilles tendon rupture in elite male athletes: a scoping review. *Br J Sports Med*. 2022;56(9):515-520.
 27. Leino O, Keskinen H, Laaksonen I, et al. Incidence and treatment trends of Achilles tendon ruptures in Finland: a nationwide study. *Orthop J Sports Med*. 2022;10(11):23259671221131536.
 28. Lerch TD, Schwinghammer A, Schmaranzer F, et al. Return to sport and patient satisfaction at 5-year follow-up after nonoperative treatment for acute Achilles tendon rupture. *Foot Ankle Int*. 2020;41(7):784-792.
 29. Lieber RL, Roberts TJ, Blemker SS, Lee SSM, Herzog W. Skeletal muscle mechanics, energetics and plasticity. *J Neuroeng Rehabil*. 2017;14(1):108.
 30. Liu X, Dai TJ, Li BL, et al. Early functional rehabilitation compared with traditional immobilization for acute Achilles tendon ruptures: a meta-analysis. *Bone Joint J*. 2021;103(6):1021-1030.
 31. Maffulli N. Rupture of the Achilles tendon. *J Bone Joint Surg Am*. 1999;81(7):1019-1036.
 32. McHugh MP, Orishimo KF, Kremenich IJ, Adelman J, Nicholas SJ. Electromyographic evidence of excessive Achilles tendon elongation during isometric contractions after Achilles tendon repair. *Orthop J Sports Med*. 2019;7(11):2325967119883357.
 33. Mullaney MJ, McHugh MP, Tyler TF, Nicholas SJ, Lee SJ. Weakness in end-range plantar flexion after Achilles tendon repair. *Am J Sports Med*. 2006;34(7):1120-1125.
 34. Nilsson L, Thorlund JB, Kjær IL, Kazlauskas A, Christensen M. Long-term follow-up after acute Achilles tendon rupture—does treatment strategy influence functional outcomes? *Foot (Edinb)*. 2021;47:101769.
 35. Nilsson-Helander K, Thomeé R, Silbernagel KG, et al. The Achilles tendon Total Rupture Score (ATRS): development and validation. *Am J Sports Med*. 2007;35(3):421-426.
 36. Nordenholm A, Nilsson N, Senorski EH, et al. Patients with chronic Achilles tendon rupture have persistent limitations in patient-reported function and calf muscle function one year after surgical treatment—a case series. *J Exp Orthop*. 2022;9(1):15.
 37. Okoroha KR, Ussef N, Jildeh TR, et al. Comparison of tendon lengthening with traditional versus accelerated rehabilitation after Achilles tendon repair: a prospective randomized controlled trial. *Am J Sports Med*. 2020;48(7):1720-1726.
 38. Pajala A, Kangas J, Siira P, Ohtonen P, Leppilahti J. Augmented compared with nonaugmented surgical repair of a fresh total Achilles tendon rupture: a prospective randomized study. *J Bone Joint Surg Am*. 2009;91(5):1092-1100.
 39. Park HG, Youn D, Baik JM, Hwang JH. Epidemiology of Achilles tendon rupture in South Korea: claims data of the National Health Insurance Service from 2009 to 2017. *Clin Orthop Surg*. 2021;13(4):539-548.
 40. Peng WC, Chang YP, Chao YH, et al. Morphomechanical alterations in the medial gastrocnemius muscle in patients with a repaired Achilles tendon: associations with outcome measures. *Clin Biomech (Bristol, Avon)*. 2017;43:50-57.
 41. Seynnes OR, Cronin NJ. Simple Muscle Architecture Analysis (SMA): an ImageJ macro tool to automate measurements in B-mode ultrasound scans. *PLoS One*. 2020;15(2):e0229034.
 42. Sharma P, Maffulli N. Biology of tendon injury: healing, modeling and remodeling. *J Musculoskelet Neuronal Interact*. 2006;6(2):181-190.
 43. Silbernagel KG, Nilsson-Helander K, Thomeé R, Eriksson BI, Karlsson J. A new measurement of heel-rise endurance with the ability to detect functional deficits in patients with Achilles tendon rupture. *Knee Surg Sports Traumatol Arthrosc*. 2010;18(2):258-264.
 44. Silbernagel KG, Steele R, Manal K. Deficits in heel-rise height and Achilles tendon elongation occur in patients recovering from an Achilles tendon rupture. *Am J Sports Med*. 2012;40(7):1564-1571.
 45. Stäudle B, Seynnes O, Laps G, Göll F, Brüggemann GP, Albracht K. Recovery from Achilles tendon repair: a combination of postsurgery outcomes and insufficient remodeling of muscle and tendon. *Med Sci Sports Exerc*. 2021;53(7):1356-1366.
 46. Svensson RB, Couppé C, Agergaard AS, et al. Persistent functional loss following ruptured Achilles tendon is associated with reduced gastrocnemius muscle fascicle length, elongated gastrocnemius and soleus tendon, and reduced muscle cross-sectional area. *Transl Sports Med*. 2019;2(6):316-324.